Interactive Power BI Dashboard Project Proposal

V Learn Platform

1. Executive Summary

This project delivers a suite of interactive Power BI dashboards for V Learn Platform, an EdTech platform focused on improving student engagement, course completion, and learner retention.

V Learn Platform collects large volumes of student, course, instructor, assessment, and platform usage data. Much of this data was previously fragmented across enrollment records, learning activity logs, quiz and assessment results, and feedback surveys. The delivered dashboarding solution consolidates this data into five connected Power BI reports that turn raw learning and platform data into clear, actionable insights for academic, product, and leadership teams.

The dashboards enable stakeholders to:

- Monitor daily active users (DAU), average progress, and platform access patterns by region, device, and difficulty level.
- Track enrollment, completion, and drop-off across courses and categories.
- Identify high, medium, and low risk learners using an average risk score.
- Evaluate instructor performance, course ratings, NPS, and learner feedback.
- Monitor quiz attempts, average quiz scores, and pass rates by course and category.
- Create a shared, filterable view of platform health and learner success across regions and course categories.

The final solution gives V Learn Platform five role-focused Power BI dashboards — Student Engagement, Course Completion, Risk Assessment, Instructor & Content Performance, and Assessment & Quiz Performance — supporting better academic decisions, product improvements, and retention strategies.

2. Problem Statement

Background

V Learn Platform had limited visibility into daily student engagement, course progression, and drop-off behavior across its platform. While data was available from enrollment records, activity logs, and assessment systems, it was not consolidated into a single reporting layer for easy analysis.

Key challenges included:

- Limited visibility into daily active user trends and how students accessed the platform (laptop, mobile, or tablet).
- Delayed reporting on course completion, drop rate, and assessment performance.
- Lack of consistent metrics across academic, product, and instructor-facing teams.
- Limited ability to compare learner outcomes across regions, course categories, and difficulty levels.
- Difficulty identifying which learners were at high risk of disengaging or dropping a course.
- Manual reporting processes that reduced speed, accuracy, and scalability.

Without centralized dashboards, teams risked missing early warning signs of student churn, low course effectiveness, or declining engagement.

Objective

The objective of this project was to build interactive Power BI dashboards that allow V Learn Platform to monitor and analyze student engagement, course completion, learner risk, instructor performance, and assessment outcomes.

The dashboards help stakeholders answer questions such as:

- How is DAU trending over time, and how does it vary by course category?

- Which devices and difficulty levels drive the most daily active usage?
- Which instructors and courses achieve the strongest ratings, NPS, and completion rates?
- How do average quiz scores and pass rates vary by course and category?
- No of enrollments fall into high, medium, or low risk and where are they concentrated by course and region?

Scope

The dashboarding scope covers five connected Power BI reports built around V Learn Platform's core engagement, completion, risk, instructor, and assessment metrics.

Key areas include:

- Daily Active Users (DAU) by activity date, region, device type, and difficulty level.
- Average progress percentage by year and region.
- Enrollment status tracking across Completed, In Progress, Near Completion, and Dropped.
- Course completion rate and drop rate by course name and category.
- Average video completion percentage alongside completion rate and enrollment volume by category.
- Risk categorization of enrollments into high, medium, and low risk, with an average risk score.
- High-risk enrollment concentration by course name and region.
- Instructor performance based on average quiz score, course completion rate, and course rating.
- Course-level feedback rating distribution and average NPS.
- Quiz attempts, average quiz score, and quiz pass rate by course and category.
- Average session duration alongside quiz score and total enrollments by course.

3. Data Sources

Primary Data

The dashboards primarily use synthetic data created for V Learn Platform including:

Learning activity logs

- Daily active user (DAU) events by activity date
- Device type used to access the platform (laptop, mobile, tablet)
- Average session duration
- Progress percentage by student and course
- Difficulty level of accessed content

Enrollment and completion records

- Total, completed, in progress, near completion, and dropped enrollments
- Course completion rate and drop rate by course name
- Enrollment volume by course category
- Average video completion percentage

Risk and assessment databases

- Enrollment-level risk score
- Risk classification into high, medium, and low risk
- Quiz attempts and average quiz scores
- Quiz pass rate by course and category
- Assessment breakdown by activity type, including applied projects, capstones, case studies, core concepts, and guided practice

Course and instructor metadata

- Course name and category (AI & Machine Learning, Business Intelligence, Career Development, Cloud Computing, Cybersecurity, Data & Analytics, Design, Education, Marketing, Product & Strategy)
- Instructor name
- Course rating and Net Promoter Score (NPS)
- Region: Asia Pacific, Europe, Latin America, Middle East & Africa, North America

Secondary Data
Secondary sources provide additional context for performance benchmarking and qualitative insight. These include:

Student feedback surveys
- Feedback rating distribution (1 to 5 scale)
- Course-level Net Promoter Score
- Instructor and course satisfaction indicators

Historical benchmarks
- Year-over-year completion rate by category (2025 vs 2026)
- Historical average progress trends by region
- Baseline engagement and risk indicators

Instructor performance records
- Average quiz score by instructor
- Course completion rate by instructor
- Total active instructors

4. Methodology

Data Integration
The project began with the extraction, transformation, and integration of data from V Learn Platform's core systems into a centralized, dashboard-ready data model.

Key activities included:

- Identifying relevant data across activity logs, enrollment records, assessment databases, and feedback surveys.
- Cleaning and standardizing fields such as course name, category, region, instructor name, and device type.
- Creating reliable relationships between enrollment, activity, risk, and assessment tables.
- Building reusable Power Query transformations and DAX measures for dashboard consumption.
- Establishing quality checks for missing values, duplicate enrollments, and inconsistent course or category labels.
- Defining business logic for key metrics such as DAU, average progress, completion rate, drop rate, risk score, and quiz pass rate.

Data pipeline flow:

- Activity and enrollment data are extracted on a scheduled basis.
- Assessment and quiz data are loaded from the assessment database.
- Course, instructor, and category metadata are joined to activity and enrollment records.
- Feedback survey data is cleaned and mapped to course and instructor records.
- Curated tables are published to Power BI datasets.
- Dashboards refresh on a scheduled basis.

Dashboard Design
Dashboard design focused on clarity, usability, and accessibility for non-technical educators, academic managers, product teams, and executives. The solution includes five Power BI dashboards:

Student Engagement Dashboard
- DAU by activity date, with a selectable date range
- Average progress percentage by year and region
- DAU access by device type (laptop, mobile, tablet)
- DAU by difficulty level (beginner, intermediate, advanced)
- Filters for region and course category

Course Completion Dashboard

- Total, completed, and dropped enrollments, course completion rate, and drop rate
- Enrollments by completion status: completed, in progress, near completion, and dropped
- Course completion rate and drop rate by course name
- Completion rate by category across 2025, 2026, and total
- Average video completion percentage, course completion rate, and total enrollments by category

Risk Assessment Dashboard

- High, medium, and low risk enrollment counts and average risk score
- High-risk enrollments by course name
- High-risk enrollments by region
- Filters for region and course name

Instructor & Content Performance Dashboard

- Total instructors, average course rating, average NPS, average video completion, and average quiz score
- Average quiz score by instructor name
- Course completion rate by instructor name
- Average course rating by course name
- Feedback score distribution by rating

Assessment & Quiz Performance Dashboard

- Quiz attempts, average quiz score, quiz pass rate, and total courses
- Average quiz score by course name
- Quiz pass rate by category
- Average session duration, average quiz score, and total enrollments by course name
- Detail table breaking down course performance by activity type, including applied projects, capstones, case studies, core concepts, and guided practice

Design principles applied across all five dashboards:

- Clean, educator-friendly layouts with high-level KPI cards at the top of each report.
- Trend charts for date-based analysis, such as DAU by activity date.
- Donut and bar charts for categorical breakdowns, such as device type, difficulty level, and feedback rating.
- Bubble charts to compare three metrics at once, such as video completion, completion rate, and enrollment volume.
- Consistent color coding across dashboards for readability.
- Slicers for course category, region, course name, and date range on every report.

Interactivity

The dashboards include interactive features that allow users to explore data without requiring technical expertise.

Delivered interactivity includes:

- Slicers for course category, region, course name, and date range.
- A date range slider to filter DAU trends by activity date.
- Cross-filtering between visuals within each dashboard, such as filtering high-risk enrollments by course and region together.
- Comparison views between courses, categories, instructors, and regions.
- Drill-through from category-level summaries to individual course names.
- A scrollable detail table for course-level assessment breakdowns by activity type.

5. Expected Outcomes

The project delivers measurable improvements in visibility, decision-making, and learner success.

Key outcomes include:

- ✓ Interactive dashboards providing near real-time insights into student engagement and platform usage.
- ✓ Clear visibility into course completion trends and drop-off, including a overall drop rate across enrollments.
- ✓ Faster identification of at-risk enrollments, with high, medium, and low risk clearly segmented and an average risk score
- ✓ Improved course completion rates through better understanding of completion status stages, from in progress to near completion.
- ✓ Stronger alignment between academic, product, and instructor-facing teams through shared metrics.
- ✓ Reduced manual reporting effort and improved reporting consistency across five connected dashboards.
- ✓ Better measurement of instructor performance, with visibility into average quiz score, completion rate, and rating by instructor.
- ✓ Clearer understanding of assessment outcomes, including an quiz pass rate and average quiz score across courses.
- ✓ Empowered educators and product teams who can act quickly using accessible, filterable data.

Success metrics include:

- Course completion rate
- Drop rate by course and category
- Average risk score and the proportion of high-risk enrollments
- Average quiz score and quiz pass rate
- Average course rating and NPS
- Dashboard adoption among academic, product, and instructor-facing teams.

6. Tools and Technologies

The delivered solution uses a modern analytics stack suitable for EdTech reporting and stakeholder collaboration.

Power BI

- Dashboard development across five connected reports
- Interactive visualizations, including donut charts, bar charts, bubble charts, and trend lines
- Slicers and cross-filtering for role-based exploration
- Scheduled data refreshes

Power Query and DAX

- Data cleaning and transformation ahead of dashboard load
- Creation of reusable measures for DAU, average progress, completion rate, drop rate, and risk score
- Calculated fields for risk classification and completion status

Microsoft Excel

- Data extraction from enrollment, activity, and assessment sources
- Data validation and reconciliation across source tables

7. Risks and Challenges

Several risks and challenges were addressed to ensure successful delivery and adoption.

Student Data Privacy and Compliance

Because the project involves student-level data, privacy and compliance were treated as critical.

Key considerations included:

- Ensuring compliance with FERPA, GDPR, and other applicable education data privacy regulations.
- Restricting sensitive student-level detail to aggregate views such as enrollment counts and category summaries.
- Defining clear rules for who can access course-level, instructor-level, and regional data.
- Maintaining secure data transfer and storage practices.

Data Quality and Integration

The dashboards rely on accurate, consistent, and timely data.

Challenges encountered included:

- Inconsistent course naming across enrollment, activity, and assessment tables.
- Differences in how completion stages (in progress, near completion, dropped) were recorded across systems.
- Reconciling risk scores with underlying enrollment and quiz data.
- Aligning category labels across course, instructor, and feedback tables.

Mitigation steps which can be included in next phase for better standardization :

- Creating standardized metric definitions for completion rate, drop rate, and risk score.
- Building validation checks into the Power Query transformation steps.
- Reconciling data between source tables and dashboard outputs.
- Documenting known data limitations for each dashboard.

Handling Feedback and Rating Data

Feedback ratings and NPS scores required careful handling to remain comparable across courses and instructors.

Challenges included:

- Uneven volumes of feedback across courses and instructors.
- Combining a 1-to-5 feedback rating scale with a separate NPS metric.
- Ensuring feedback score distributions were not skewed by low-response courses.

Solutions applied included:

- Standardizing feedback rating buckets across the Instructor & Content Performance dashboard.
- Cross-checking NPS and feedback rating trends against course completion and quiz performance.
- Flagging courses with low feedback volume for manual review.

User Adoption and Change Management

Educator and product team adoption is essential for the dashboards to create business value.

Potential adoption barriers included:

- Stakeholders unfamiliar with Power BI slicers and cross-filtering.
- Lack of confidence interpreting risk scores and completion status stages.
- Preference for existing manual reports.
- Limited time for educators to explore five separate dashboards.

Mitigation steps included:

- Conducting stakeholder workshops during dashboard design.
- Creating a metric glossary covering DAU, risk score, completion rate, and quiz pass rate.
- Providing short training sessions for educators and academic managers.
- Collecting feedback after launch and iterating on layout and filters.

8. Conclusion

This project equips V Learn Platform with a Power BI dashboarding solution that transforms fragmented student and platform data into actionable intelligence. Across five connected dashboards — Student Engagement, Course Completion, Risk Assessment, Instructor & Content Performance, and Assessment & Quiz Performance — V Learn Platform can monitor engagement, completion, risk, instructor effectiveness, and assessment performance in a centralized, user-friendly way.

The delivered dashboards help academic teams identify at-risk learners earlier, help product teams understand where and why students drop off, and give leadership reliable visibility into platform performance. Most importantly, the project helps V Learn Platform build a stronger data-driven educational culture where decisions are guided by evidence, learner insight and metrics.